Review: Properties of Integrals

- $\int_a^a f(x)\,dx = 0$
- $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
- $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$
- $\int_a^b (f(x) \pm g(x))\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$
- $\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$
- If $f(x) \le g(x)$ for $a \le x \le b$, then $\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$
- If $m \le f(x) \le M$ for $a \le x \le b$, then $m(b-a) \le \int_a^b f(x)\,dx \le M(b-a)$

Antiderivatives and Indefinite Integrals

Question Suppose $F(x)$ is an antiderivative for $f(x)$. What does this mean?

Let's introduce some new notation. Let $\int f(x)\,dx$ represent all possible antiderivatives for $f(x)$. We call this the **indefinite** integral.

Questions If $F(x)$ is an antiderivative for $f(x)$, why is $F(x) + C$ also an antiderivative, where C is any constant? (Hint: Use the question above. How do you check if something is an antiderivative of something else?)

How many possible antiderivatives does a function have and how are they connected?

Basic Antiderivatives

Fill in the following basic integration formulas. Here, k and n are constants.

- $\int k\,dx =$ _____
- $\int e^x\,dx =$ _____
- $\int x^n\,dx =$ _____ for $n \neq -1$
- $\int \sin x\,dx =$ _____ (Hint: what function's derivative is $\sin x$?)
- $\int \cos x\,dx =$ _____

Integral of $\frac{1}{x}$

Why do we need to stipulate that $n \neq -1$ for $\int x^n$ above?

If $x > 0$, what is $\int \frac{1}{x}\,dx$? Why must we stipulate $x > 0$ here?

If $x < 0$, show that $\ln(-x)$ is an antiderivative for $\frac{1}{x}$. Why is $\ln(-x)$ defined in this case?

Explain why this shows that $\int \frac{1}{x}\,dx = \ln|x| + C$.

We also have the properties

$$\int (f(x) + g(x))\,dx = \int f(x)\,dx + \int g(x)\,dx$$

and

$$\int cf(x)\,dx = c\int f(x)\,dx.$$

It's worth your time to think about why these are true.

Definite vs Indefinite Integrals

We previously defined the *definite* integral of a function $f(x)$ between $x = a$ and $x = b$:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(a + i\Delta x)\Delta x, \text{ where } \Delta x = \frac{b-a}{n}.$$

Write the differences between $\displaystyle\int_a^b f(x)\,dx$ and $\displaystyle\int f(x)\,dx$ in each of the following categories:

- Computing $\displaystyle\int_a^b f(x)\,dx$ for a given function and numbers a and b in its domain gives a _____. Computing $\displaystyle\int f(x)\,dx$ gives a _____.
- When you compute $\displaystyle\int_a^b f(x)\,dx$ for a given function and numbers a and b in its domain, what does your result mean in relation to the (graph of) the function?
- If you computed $\displaystyle\int f(x)\,dx$, how could you use your answer to compute the same quantity as in the previous bullet point? Specifically, what theorem allows you to do so?
- We defined the definite integral $\displaystyle\int_a^b f(x)\,dx$ as a limit of _____. The indefinite integral $\displaystyle\int f(x)\,dx$ is the set of all _____ of $f(x)$. They are connected, but distinct!

Questions!

1. A student wrote that $\int \cos(2x)\ dx = \sin(2x) + C$. Check their answer by taking a derivative, and if needed, correct it and check your revised answer. □Regrade

2. To find $\int e^{2x}\ dx$, you might first guess that an antiderivative is $e^{2x} + C$. But when we check, we get $\frac{d}{dx}e^{2x} = 2e^{2x}$. So this is incorrect. How could you use this check to find the correct antiderivative? □Regrade

3. Evaluate $\int \sin\left(-\frac{x}{2}\right) + e^{17x}\,dx$ and carefully check your answer. If needed, use similar ideas to the previous questions to correct your work. □Regrade

4. Does a similar correction procedure work for $\int e^{-x^2}\,dx$? A first guess might be e^{-x^2}. But $\frac{d}{dx}e^{-x^2} = -2xe^{-x^2}$. So is $\frac{e^{-x^2}}{-2x}$ an antiderivative of e^{-x^2}? Check by using differentiation. Why doesn't this similar correction work here? □Regrade

5. ☐Regrade (a) What is $\displaystyle\int \frac{1}{1+x^2}\,dx$? (Hint: for this and the next one, look back at the differentiating inverse trig functions worksheet.)

(b) What is $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx$?

(c) Let's look at $\displaystyle\int -\frac{1}{\sqrt{1-x^2}}\,dx$ in two different ways.

i. First, compute this integral using your answer to part (b).

ii. Looking back at the differentiating inverse trig functions worksheet, what function has derivative $-\frac{1}{\sqrt{1-x^2}}\,dx$? What does that say that $\displaystyle -\frac{1}{\sqrt{1-x^2}}\,dx$ is?

iii. Look back at the bottom of page 1. What must be true about the relationship between $-\arcsin x$ and $\arccos x$? Draw their graphs to show it's true.

Summary

For now, we can only find antiderivatives of functions that we can recognize as the derivative of something else. For example, we know that the derivative of e^{kx} is ke^{kx}, so we can conclude that the function $\frac{1}{k}e^{kx}$ has derivative e^{kx}. Therefore, $\frac{1}{k}e^{kx}$ is an antiderivative of e^{kx}:

$$\begin{aligned} \frac{d}{dx}e^{kx} &= ke^{kx} \\ \Rightarrow \frac{1}{k}\frac{d}{dx}e^{kx} &= e^{kx} \\ \Rightarrow \frac{d}{dx}\frac{1}{k}e^{kx} &= e^{kx} \\ \Rightarrow \int e^{kx}\,dx &= \frac{1}{k}e^{kx} + C. \end{aligned}$$

However, if we can't easily find a function whose derivative is what we're given, we don't know how to find an antiderivative. Next week, we'll figure out how to reverse the product rule and chain rule to add to our library of antiderivatives.

Having said that, there are many functions for which it is very hard (or even impossible– in the case of e^{-x^2}, for example) to write down 'nice' antiderivatives. What exactly 'nice' means, and how we can get around that, at least for theoretical applications, will be covered at the end of Week 9.